National Higher School of Autonomous Systems Technology
Preparatory Cycle, 2nd Year
February 2026

Strength of Materials
Tutorial Series 1

Exercise 1

Classify each of the following structures represented in Figure 1 as statically determinate or statically indeterminate. Justify your answers.

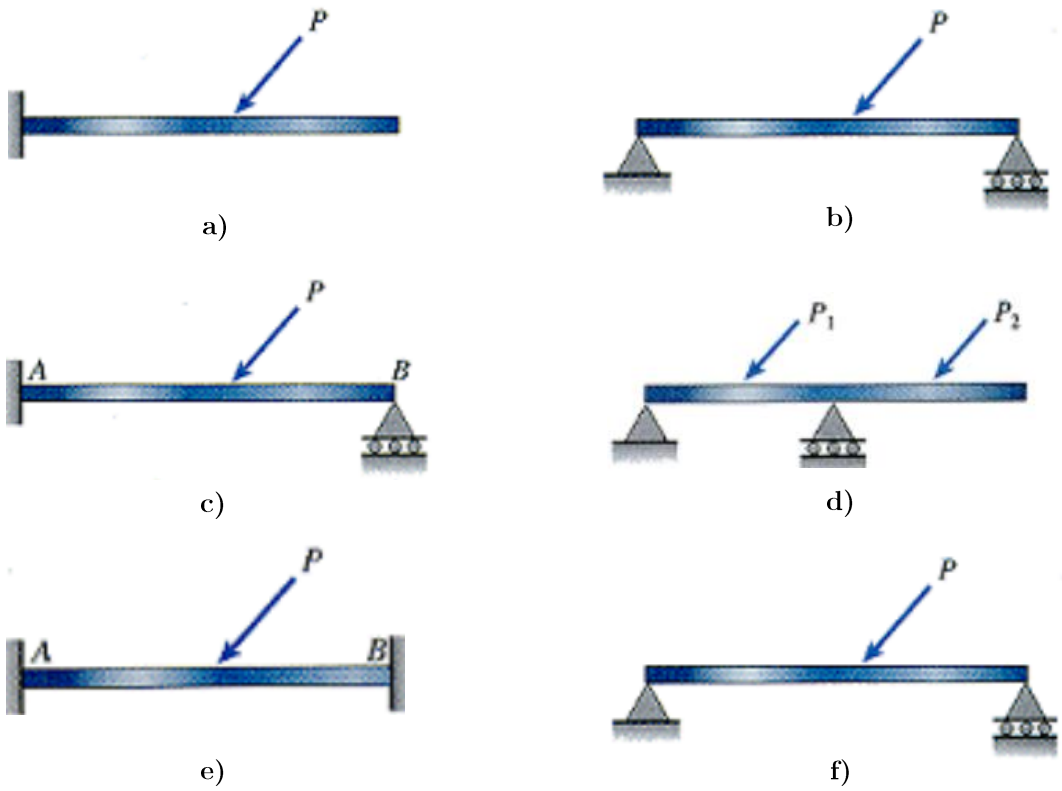

Figure 1

Exercise 2

A structure is composed of two rods : a horizontal rod AC and an inclined rod BC, which forms an angle α with the vertical. The two rods are connected by a hinge at point C (Figure 2).

A load P is applied at point C. Neglecting the weight of the rods, determine the reactions at the supports A and B.

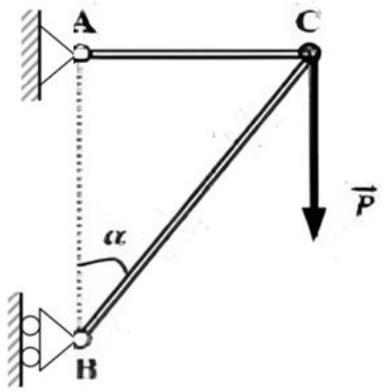

Figure 2

Exercise 03

Determine the support reactions of the beam shown in Figure 3. The weight of the beam is negligible.

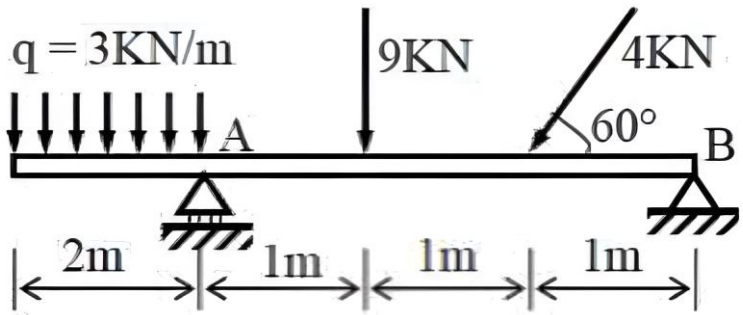

Figure 3